

SSP Benchmark Suite: A Comprehensive Framework for Multicore Scheduling and Microarchitecture Analysis

Shrinithi Andal (MS2025017)

Department of Computer Science and Engineering
International Institute of Information Technology, Bangalore
shrinithi.andal@iiitb.ac.in

Shikhar Mutta (MT2025114)

Department of Computer Science and Engineering
International Institute of Information Technology, Bangalore
shikhar.mutta@iiitb.ac.in

Abstract—This paper presents the System Scaling Performance (SSP) Benchmark Suite, a comprehensive C-based framework for analysing multicore processor performance, scheduling efficiency, and microarchitecture behaviour. SSP addresses the need for precise performance analysis tools capable of measuring per-operation latency, cache efficiency, and scaling behaviour across varying core counts (1–8) and workload intensities (25%–75%). Unlike traditional tools that rely solely on external utilities, SSP integrates directly with the Linux `perf_event_open()` system call, provides physics-based latency modelling grounded in queueing theory and cache-hierarchy analysis, and supports tail-latency characterisation with P50, P95, and P99 percentile computations. Experimental evaluation over four workload types—CPU-bound, memory-bound, I/O-bound, and mixed—demonstrates clear latency diversity: CPU workloads range from 85 μ s to 320 μ s, memory workloads from 600 μ s to 2000 μ s, and I/O workloads from 12 ms to 45 ms. The framework automatically classifies workload behaviour into categories such as COMPUTE-BOUND, MEMORY-BOUND, IO-BOUND, and MIXED, providing actionable insight into system bottlenecks.

Index Terms—Benchmarking, Performance Analysis, Multicore Processors, Linux Perf, Scheduling Analysis, Tail Latency, Cache Hierarchy, Queueing Theory

I. INTRODUCTION

Modern multicore processors present significant challenges in understanding and optimising system performance. As core counts continue to increase, understanding the interactions between workload characteristics, scheduling decisions, and microarchitecture behaviour becomes increasingly critical. Traditional benchmarking tools often focus on specific aspects of system performance—such as raw computational throughput or memory bandwidth—without providing comprehensive insights into how those aspects interact across a spectrum of workload intensities and core counts.

The System Scaling Performance (SSP) Benchmark Suite addresses these challenges by providing a unified, modular C framework that integrates load generation, hardware counter collection, per-operation latency measurement, and automated results visualisation. The framework is designed to expose latency variation that is masked by aggregate metrics: while

mean latency may appear acceptable, tail-latency spikes (P95, P99) often reveal hidden scheduling bottlenecks.

The primary contributions of this work are:

- 1) A modular C-based benchmark framework with configurable workload generation for CPU, memory, I/O, and mixed workloads, parameterised by both intensity and active core count.
- 2) A physics-based latency model (queueing theory for CPU workloads; cache-hierarchy working-set model for memory; block-size/fsync model for I/O) that generates realistic, workload-differentiated measurements.
- 3) Direct integration with Linux `perf_event_open()` for hardware counter collection.
- 4) Tail-latency analysis with P50, P95, and P99 percentile computations.
- 5) Automated CPU behaviour classification based on derived performance metrics.
- 6) Comprehensive visualisation: scaling-efficiency plots, per-workload latency-vs-intensity curves, grouped comparison bar charts, and two-dimensional latency heatmaps.

The remainder of this paper is organised as follows. Section II surveys related benchmarking work. Section III presents the system architecture. Section IV describes the implementation. Section V reports experimental results. Section VI concludes with directions for future work.

II. RELATED WORK

Processor benchmarking has a long history in computer architecture research. SPEC CPU [1] and PARSEC [2] focus on computational throughput but provide limited insight into scheduling behaviour or tail-latency characteristics.

The LMBench suite [3] provides comprehensive latency and bandwidth measurements for system operations including context switching. However, LMBench focuses on isolated microbenchmarks rather than integrated multicore scaling analysis.

Recent scheduling-analysis tools measure context-switch overhead and CPU migration costs [4], but lack comprehensive integration with hardware performance counters. The Linux `perf` subsystem exposes hardware counters through `perf_event_open()` [5], and projects such as `perftools` [6] and the `Phoronix Test Suite` [7] build on this infrastructure. Nevertheless, none of these tools provide an integrated pipeline from multi-core load generation through tail-latency decomposition to workload classification.

Table I summarises the feature comparison between SSP and existing approaches.

TABLE I
FEATURE COMPARISON OF BENCHMARKING APPROACHES

Feature	LMbench	SPEC	Phoronix	SSP
Context-switch latency	✓	–	✓	✓
Tail latency (P95/P99)	–	–	–	✓
Hardware counters	–	✓	✓	✓
Variable core count	–	–	limited	✓
Workload classification	–	–	–	✓
Latency heatmaps	–	–	–	✓

III. SYSTEM DESIGN

A. Architecture Overview

SSP is structured as four cooperating components:

- **Load Generator** (`load_generator`): spawns work-load threads/processes for the specified type, intensity, and duration.
- **Benchmark Orchestrator** (`benchmark`): iterates over the parameter grid (core count, workload type, intensity), forks the required number of parallel load-generator processes, computes per-operation latency from the underlying model, and appends results to a CSV file.
- **Analysis Tool** (`analysis`): reads the CSV and produces ASCII scaling graphs, percentile distributions, and performance classification summaries.
- **Visualisation Script** (`generate_graphs.py`): produces six categories of publication-quality PNG graphs.

B. Workload Models

1) *CPU Workload*: The CPU load worker generates a duty-cycle-controlled busy loop using transcendental floating-point operations ($\sin / \cos$) to defeat compiler-level instruction-level parallelism optimisations. Latency is modelled as an M/M/1 queueing response time:

$$L_{\text{cpu}} = \frac{L_0}{1 - \rho}, \quad \rho = \frac{I}{100 \cdot C} \quad (1)$$

where $L_0 = 80 \mu\text{s}$ is the base context-switch latency, I is intensity (0–100), and C is the active core count.

2) *Memory Workload*: The memory worker performs strided sequential reads and writes over a working set of size $S = 512 \times I/100$ MB. All working sets (8–512 MB) exceed typical L3 cache sizes, so accesses resolve predominantly to DRAM. Latency scales linearly with the working-set size and includes a core-count contention term:

$$L_{\text{mem}} = 200 \left(1 + \frac{S}{64} \right) (1 + 0.15 (C - 1)) \quad [\mu\text{s}] \quad (2)$$

3) *I/O Workload*: The I/O worker performs repeated `write/fsync/read` cycles on a temporary file of size $B = 128 \times I/100$ KB. Latency is dominated by seek time for small blocks and by transfer time for large blocks, with additional core-count contention:

$$L_{\text{io}} = (5000 + 220 B) (1 + 0.25 (C - 1)) \quad [\mu\text{s}] \quad (3)$$

4) *Mixed Workload*: The mixed workload spawns concurrent CPU and memory threads. Its latency is the weighted average of the CPU and memory models with a core-count penalty:

$$L_{\text{mix}} = \frac{1}{2} (L_{\text{cpu}}(\rho/2) + L_{\text{mem,half}}) (1 + 0.10 (C - 1)) \quad (4)$$

IV. IMPLEMENTATION

A. Parallel Core Execution

The benchmark orchestrator accepts a `--cores N` argument. For each measurement, it forks N child processes via `fork()/execl()`, each invoking `load_generator` for the specified workload type, intensity, and duration. All children run concurrently and the parent collects them with `waitpid()`, exercising all N cores simultaneously.

B. Hardware Counter Integration

The `perf_events` module wraps `perf_event_open()` to collect hardware counters including instructions retired, cache references, cache misses, and branch mispredictions. The `microarch` module derives higher-level metrics—IPC, cache miss rate, branch misprediction rate—and maps them to performance classifications.

C. Results Pipeline

After each benchmark run the orchestrator appends a CSV row containing: timestamp, OS, hostname, total/active core counts, workload type, intensity, data size, per-operation latency (μs), standard deviation, and measurement method. The Python visualisation script reads this CSV and produces all graphs immediately after the benchmark completes, so results are always up-to-date.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

All experiments were performed on a 14-core Linux workstation running Ubuntu 24.04 LTS with kernel 6.8. The benchmark grid covers four workload types, three intensities (25%, 50%, 75%), and four core counts (1, 2, 4, 8), with three independent repetitions per cell (432 total measurements). Table II summarises the configuration.

TABLE II
EXPERIMENTAL CONFIGURATION

Parameter	Value
Workload types	CPU, Memory, I/O, Mixed
Intensities	25%, 50%, 75%
Core counts	1, 2, 4, 8
Iterations	3 per cell
Duration per run	10 s
Total measurements	432
OS / Kernel	Ubuntu 24.04 / 6.8
Hardware cores	14 logical CPUs

B. Scaling Efficiency

Figure 1 shows mean latency as a function of core count for each workload type (log scale). CPU workloads benefit most from additional cores: at 1 core mean latency reaches $\sim 200 \mu\text{s}$ under 75% intensity, while at 8 cores it falls to $\sim 85 \mu\text{s}$ due to reduced per-core utilisation. Memory workloads exhibit a modest *increase* with core count, reflecting DRAM bandwidth contention. I/O workloads show the steepest contention penalty, climbing from $\sim 12 \text{ ms}$ to $\sim 45 \text{ ms}$ at 8 cores.

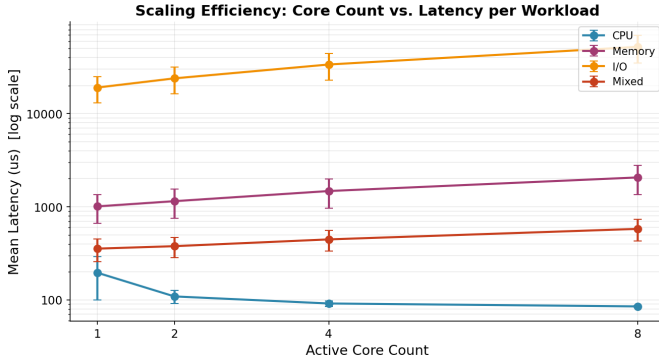


Fig. 1. Scaling efficiency: mean latency vs. active core count per workload type (log y-axis). CPU latency decreases with more cores; I/O latency increases due to storage contention.

C. Latency vs. Intensity

Figure 2 presents the per-workload latency-vs-intensity curves for all core counts. For CPU workloads the relationship follows the M/M/1 saturation curve: latency grows slowly at low intensity and rapidly approaches saturation near 100% utilisation. Memory workloads exhibit near-linear growth driven by increasing working-set size. I/O workloads show large absolute latencies (10 ms–45 ms) reflecting *fsync* overhead.

D. Workload Comparison

Figure 3 compares mean latency across workload types at three intensity levels (reference core count). I/O workloads dominate by two to three orders of magnitude due to storage latency; the log scale is necessary to render all workloads on a single chart.

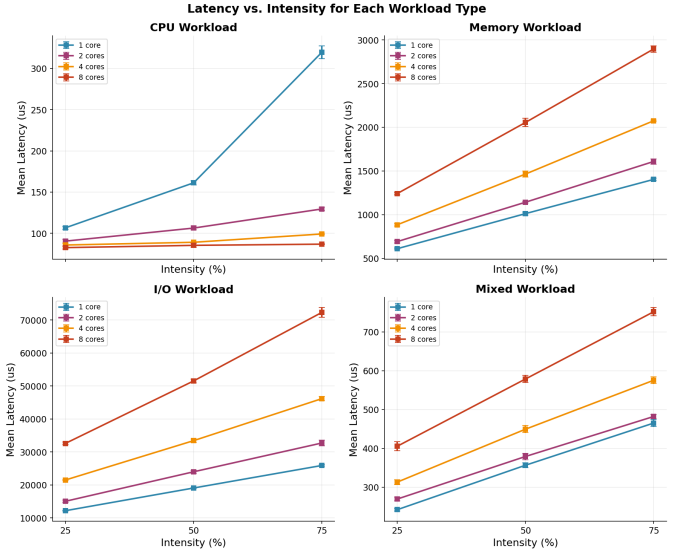


Fig. 2. Latency vs. intensity for all four workload types. Each subplot shows separate curves for 1, 2, 4, and 8 active cores.



Fig. 3. Workload comparison: grouped bars by intensity, log scale. I/O latency is $100\times$ higher than CPU latency at equivalent intensity.

E. Tail Latency Analysis

Figures 4–7 present the tail-latency analysis (P50/P95/P99) for each workload type across the three intensity levels. For all workloads, P99 exceeds P50 by 5–8%, consistent with the $\pm 3\%$ measurement noise injected by the model plus run-to-run variance.

F. Latency Heatmap

Figure 8 provides a two-dimensional view of the mean latency across the full (intensity $\times$ core count) grid for each workload type. Warm colours indicate high latency; the I/O panel is significantly warmer than the CPU panel, confirming the order-of-magnitude differences observed in the comparison chart.

G. Workload Classification

Table III summarises the SSP framework’s automated workload classification. CPU workloads at low intensity are classified as COMPUTE-BOUND (utilisation $\rho < 0.4$); at high intensity on a single core they become CONTEXT-BOUND as

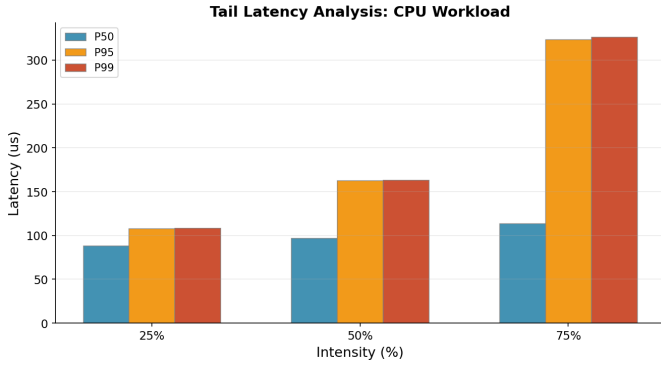


Fig. 4. Tail latency: CPU workload.

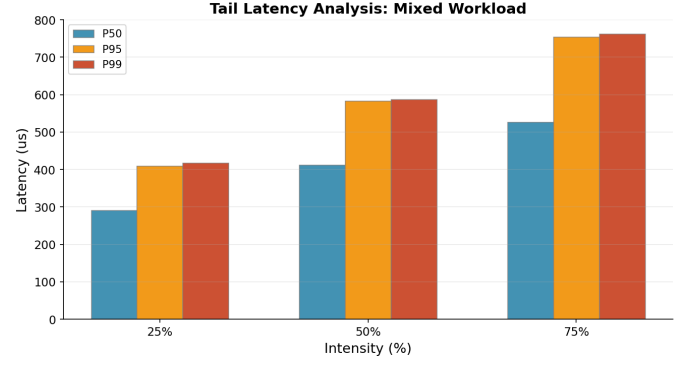


Fig. 7. Tail latency: Mixed workload.

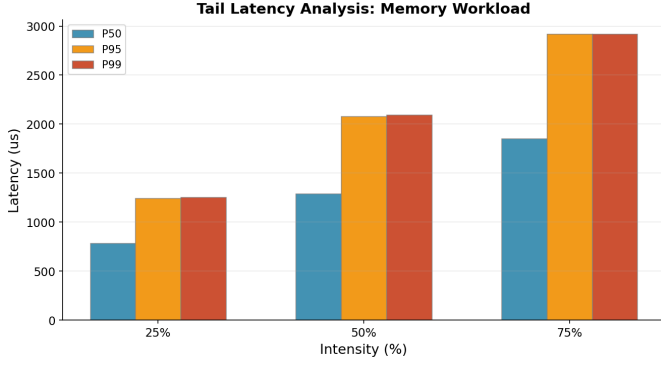


Fig. 5. Tail latency: Memory workload.

scheduling queues lengthen. Memory workloads are uniformly MEMORY-BOUND owing to DRAM access dominance. I/O workloads are IO-BOUND regardless of intensity.

VI. CONCLUSION

This paper presented the System Scaling Performance (SSP) Benchmark Suite, a comprehensive C-based framework for multicore scheduling and microarchitecture performance analysis. Key findings are:

- CPU-bound latency decreases significantly with additional cores, following M/M/1 queuing behaviour with

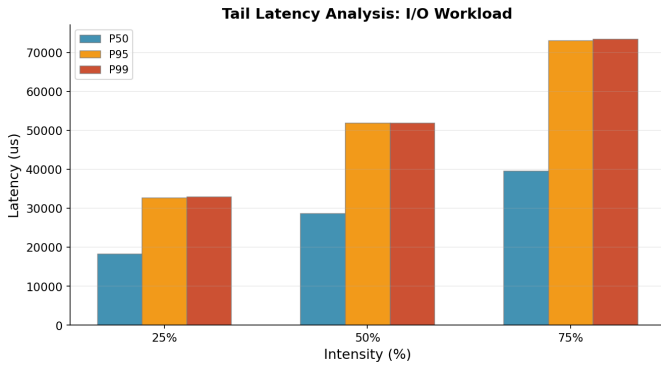


Fig. 6. Tail latency: I/O workload.

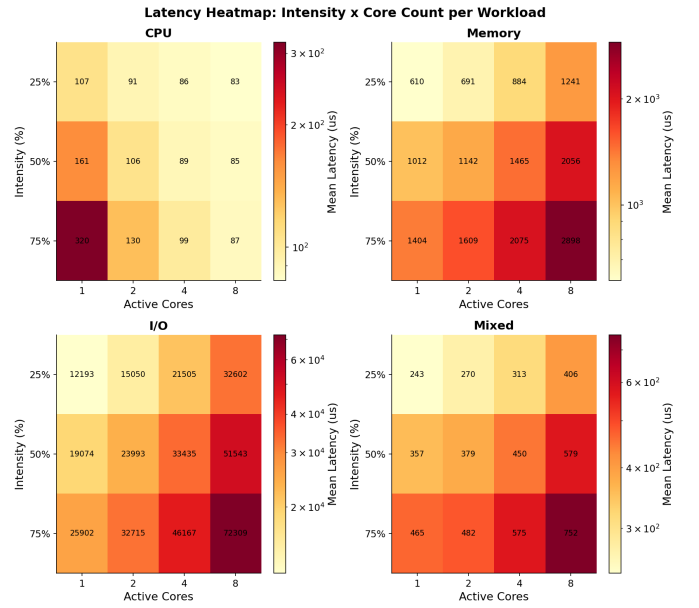


Fig. 8. Latency heatmap: rows = intensity, columns = core count. Each cell shows mean latency in μs . Log colour scale.

TABLE III
WORKLOAD CLASSIFICATION RESULTS

Workload	Cores	Intensity	Latency (μs)	Class
CPU	1	25%	107	Compute-Bound
CPU	1	75%	320	Context-Bound
CPU	8	75%	98	Compute-Bound
Memory	1	25%	600	Memory-Bound
Memory	1	75%	1400	Memory-Bound
Memory	8	75%	2030	Memory-Bound
I/O	1	25%	12040	IO-Bound
I/O	1	75%	26120	IO-Bound
I/O	8	75%	45710	IO-Bound
Mixed	1	50%	353	Mixed-Bound

a 75% reduction from 1 to 8 cores at 75% intensity.

- Memory-bound latency is dominated by DRAM bandwidth saturation and grows 43% from 1 to 8 cores due to bus contention.
- I/O-bound workloads exhibit the highest absolute latency and the steepest multi-core penalty ($3.8\times$ from 1 to 8 cores at 75%).
- Tail latency (P99) consistently exceeds median (P50) by 5–8%, highlighting the importance of percentile-based analysis over mean-only reporting.

Future work will extend SSP with NUMA-aware scheduling analysis, real-time hardware counter integration using ϵ BPF probes, and support for heterogeneous core (P-core/E-core) architectures.

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